Class 11 Practice Sheet

Nov 4, 2026 · @Laura

Directions

These are the same 30 exercises as the Class 11 app, in the same order, so you can work on paper and follow along in class. The answer key is at the end.

Name: _____ Date: _____

Part 1: Names, predicates, and quantity words

Name the expression

For each sentence, say what kind of expression the **bold** phrase is: proper name, definite description, pronoun, quantity expression, or none of these.

1. Ana's only sister is a pilot.
Kind of expression: _____

2. Nobody finished the puzzle.
Kind of expression: _____

3. Priya said **she** would come early.
Kind of expression: _____

4. Leo's cousin lives in Ohio.
Kind of expression: _____

5. How many places does the predicate 'x sits between y and z' have?
Number of places: _____

Quick check

Circle the best answer.

6. Which of these is a singular term?

A. 'the tallest mountain in Peru'
B. 'nobody'
C. 'is taller than'
D. 'everyone'

7. Start with ‘Lisbon is west of Madrid’. Which can you NOT get by deleting singular terms?

A. ‘x is west of y’
B. ‘Lisbon is x of Madrid’
C. ‘x is west of Madrid’
D. ‘Lisbon is west of x’

8. ‘Every member of the chess club plays online. Mia is a member of the chess club. So Mia plays online.’ Why does sentential logic call this valid argument invalid?

A. Each sentence becomes its own unrelated letter, so the shared name and predicates disappear
B. Sentential logic can’t handle three sentences
C. Because it really is invalid
D. Sentential logic only works when the premises are true

9. Which singular term does NOT refer to anything?

A. ‘the inventor of the telephone’
B. ‘Toronto’
C. ‘Frida Kahlo’
D. ‘the present king of Antarctica’

Part 2: Reading formulas

Read a formula

For each formula: (a) Is it atomic, truth-functional, or quantified? (b) What is its main operator? (c) Which variables are free? (d) Is it a sentence?

10. (∀x)(Fx ∨ Gy)
(a) _____ (b) _____ (c) _____ (d) yes / no

11. (∃y)Fy & Gx
(a) _____ (b) _____ (c) _____ (d) yes / no

12. ~(∀x)(Fx ⊃ (∃y)Rxy)
(a) _____ (b) _____ (c) _____ (d) yes / no

Substitution instances

Circle every substitution instance of the sentence. If it has none, write “none”.

13. (∃x)(Hx ⊃ (∃z)Hz)

a) Hc ⊃ (∃z)Hz
b) Ha ⊃ (∃z)Hz
c) (∃x)(Hc ⊃ (∃z)Hz)

14. (∃y)(∃x)(Hx ⊃ Hy)

a) (∃y)(∃x)(Hx ⊃ Ha)
b) Hb ⊃ Ha
c) (∃x)(Hx ⊃ Hb)
d) (∃x)(Hx ⊃ Ha)

15. (∀y)((∀z)Rcz & Gy)

a) (∀z)Rcz & Gd
b) (∀z)Rcz & Gc
c) (∀y)((∀z)Rcz & Gd)

Quick check

Circle the best answer.

16. Which symbol is an individual constant?

A. x
B. m
C. w
D. M

17. Which is NOT a formula?

A. Lbz
B. B
C. (∃b)Hb
D. (∃x)Hx

18. Which of these is NOT a formula?

A. Lxy
B. (∃y)Gx
C. (∃x)(Fx & Gy)
D. ~(∀z)Hz

19. Which of these is NOT a formula?

A. $(\forall x)(Fx \supset (\exists x)Gx)$
B. $Fa \supset (\forall y)Gy$
C. $(\forall x)(Fx \supset (\exists y)Gxy)$
D. $(\exists z)(Fz \equiv Ga)$

20. $(\exists y)(Gy \vee Hb)$ and $(\exists y)Gy \vee Hb$ start with the same symbol. What is the difference?

A. The second is not a formula
B. The first is quantified; the second is truth-functional with main operator $\vee$
C. The first is not a sentence
D. None, they are the same sentence

Part 3: All, no, and some

Symbolize a sentence

For each sentence: (a) name its form (A, E, I, O, or all and only), then (b) symbolize it with the key given.

21. Every shark is a predator.
UD: animals. Sx: x is a shark. Px: x is a predator.
(a) Form: _____ (b) _____

22. No doctors are runners.
UD: people. Dx: x is a doctor. Rx: x is a runner.
(a) Form: _____ (b) _____

23. At least one poet is a teacher.
UD: people. Px: x is a poet. Tx: x is a teacher.
(a) Form: _____ (b) _____

24. Not all students are athletes.
UD: people. Sx: x is a student. Ax: x is an athlete.
(a) Form: _____ (b) _____

25. Only plants are ferns.
UD: living things. Px: x is a plant. Fx: x is a fern.
(a) Form: _____ (b) _____

Quick check

Circle the best answer.

26. Which sentence is the contradictory of ‘Some students passed’?

A. ‘Some students did not pass.’
B. ‘No students passed.’
C. ‘All students passed.’
D. ‘Only students passed.’

27. UD: the students in PHIL 270. Px: x passed. Symbolize ‘Everyone passed.’

A. (∀x)(Sx ⊃ Px)
B. (∃x)Px
C. (∀x)Px
D. Pe

28. Every student passed. Is ‘Some students passed’ true?

A. Yes. ‘Some’ means at least one
B. It can’t be decided
C. No. ‘Some’ means more than one
D. No. ‘Some’ implies ‘not all’

29. There are no dragons in the UD. Dx: x is a dragon. Wx: x has wings. Which is true?

A. None of them
B. (∃x)(Dx & Wx)
C. (∀x)(Dx ⊃ Wx)
D. (∃x)(Dx & ~Wx)

30. Which is equivalent to (∀x)(Fx ⊃ ~Gx), ‘No Fs are G’?

A. ~(∃x)(Fx & Gx)
B. (∀x)(~Fx ⊃ Gx)
C. ~(∀x)(Fx ⊃ Gx)
D. (∃x)(Fx & ~Gx)

Answer key

Part 1

1. Definite description. ‘Only’ limits it to at most one person, so it is a definite description.

2. Quantity expression. It sits where a name could go, but it names no one. It says there is no person who finished.

3. Pronoun. ‘She’ stands in for ‘Priya’ and refers to the same person, so it works as a singular term.

4. None of these. Leo may have many cousins, so the phrase does not pick out one person.

5. 3 places. The gaps are x, y, and z.

6. A. A singular term picks out one thing. ‘Everyone’ and ‘nobody’ are quantity words; ‘is taller than’ is a predicate.

7. B. Only singular terms can become gaps. ‘West’ is part of the predicate.

8. A. In sentential logic it is just C, M, so P. The argument is valid because of the shared name ‘Mia’ and the shared predicates, which only predicate logic can show.

9. D. Antarctica has no king. In our symbolizations we assume every constant refers to something.

Part 2

10. (a) Quantified. (b) (∀x). (c) y. (d) No, because y is free.

11. (a) Truth-functional. (b) &. (c) x. (d) No, because x is free.

12. (a) Truth-functional. (b) ~. (c) None. (d) Yes, it has no free variables.

13. a and b. Not c: the initial quantifier has to be dropped, and keeping it leaves a quantifier with no x to bind.

14. c and d. Not a: the initial quantifier has to be dropped. Not b: it drops two quantifiers, and only the initial one goes.

15. a and b. Any constant works if it replaces every y. Not c: the initial quantifier has to be dropped.

16. B. Constants are lowercase a through v. Lowercase w, x, y, z are variables, and capitals are predicates.

17. C. Quantifiers take variables, not constants. The capital letter B on its own is a sentence letter, which is an atomic formula.

18. B. A quantifier has to bind something: (∃y) can only go in front of a formula that contains y. Gx has no y.

19. A. You can’t put an x-quantifier in front of a formula that already has one. (Fx ⊃ (∃x)Gx) already contains (∃x).

20. B. Outer parentheses may be dropped, so a formula that starts with a quantifier is not always quantified. In the second, (∃y) covers only Gy.

Part 3

21. (a) A (all). (b) (∀x)(Sx ⊃ Px)

22. (a) E (no). (b) (∀x)(Dx ⊃ ~Rx), or equivalently ~(∃x)(Dx & Rx)

23. (a) I (some). (b) (∃x)(Px & Tx)

24. (a) O (some … not). (b) (∃x)(Sx & ~Ax), or equivalently ~(∀x)(Sx ⊃ Ax)

25. (a) A (all). (b) (∀x)(Fx ⊃ Px). ‘Only plants are ferns’ means ‘All ferns are plants’, so the term after ‘only’ goes in the consequent.

26. B. Contradictories always have opposite truth values. E and I are contradictories, and so are A and O.

27. C. When the UD is only students, no ‘student’ predicate is needed.

28. A. In logic, ‘some’ means at least one and does not imply ‘not all’.

29. C. When nothing is a dragon, every conditional ‘if x is a dragon…’ has a false antecedent, so the universal sentence is true and the existential ones are false.

30. A. No Fs are G means there is no F that is G.